

What features do AI detectors focus on? (Frequency, Upsampling, Noise, Texture, Semantic, etc.)

- **Frequency-domain artifacts:**
 - Generated images often deviate from the smooth/natural frequency decay of real photos. (*frequency domain shows how image details are distributed across different levels of sharpness or detail.*)
 - They may have excess high-frequency energy (too much fine detail), periodic spikes (repeated patterns), or altered radial patterns.
 - Diffusion models (**newer type of image generator**, such as **Stable Diffusion or DALL-E 3**, that starts with random noise and refines it step-by-step into a realistic image, different from older GAN models that create an image in one shot) reduced obvious spectral peaks but still show subtle frequency differences that detectors can spot.
- **Upsampling patterns:**
 - Upsampling is the process of increasing an image's resolution by generating additional pixels to enhance detail and sharpness.
 - When AI models upsample, small repeating patterns or pixel correlations (like faint checkerboards or edge artifacts) can appear.
 - Even with newer diffusion models (*build images step-by-step from noise*), detectors still pick up on these subtle pixel faults.
- **Noise signatures:**
 - Real cameras produce consistent sensor noise (PRNU - Photo Response Non-Uniformity, brightness variations caused by small imperfections in each camera sensor) and demosaicing traces (subtle repeating color patterns created when the camera combines red, green, and blue pixel data into a full-color image).
 - AI-generated images usually lack these natural camera traces.
 - Detectors analyze the fine pixel-level noise in an image to check for these missing or fake camera signals.
- **Texture inconsistencies:**
 - AI often struggles with natural textures such as skin, hair, or fabric. Generated textures might look overly smooth, too regular, etc.
 - Texture-based detectors focus on these surface-level inconsistencies rather than overall image meaning.
- **Semantic anomalies:**
 - Logical or physical mistakes (strange hands, inconsistent shadows, warped text, etc.) often appear in AI-generated images.
 - Detectors using semantic reasoning catch these **content-based clues**, looking beyond just pixel and texture analysis.

Which features have remained consistent and which have evolved?

- **Consistent features:**
 - **Noise-based cues:** Real camera noise (PRNU/demosaicing) is still one of the most reliable differences between real and AI-generated images.

- **Frequency-domain analysis:** Using deviations in frequency continue to work well across different models.
- **Texture-level features:** Texture irregularities and pixel relationships remain strong general indicators of AI generation.
- **Evolved features:**
 - **Hybrid spatial-frequency models:** Combining pixel-space and frequency-space analysis has improved detection against new diffusion models.
 - **Semantic anomaly detection:** Modern detectors now use AI reasoning (like Vision-Language Models) to identify *logic errors* in content.
 - **Multi-scale noise consistency checks:** Recent methods compare noise patterns at multiple resolutions to catch more advanced AI-generated noise.
- **Less effective or outdated cues:**
 - **Checkerboard artifacts:** Common in older GANs but mostly gone in newer models.
 - **Single-type detectors:** Methods that only analyze pixels or one domain don't work well anymore (*must combine frequency + pixel analysis*)
 - **Model-specific training:** Detectors focused on one generator (like StyleGAN) fail on newer types (modern ones use general/flexible features)

What detection patterns generalize well across different AI image generators?

- Model real-image frequency behavior and mark deviations from it.
- Compare PRNU and noise patterns at different scales (pixel-level to picture-level) to detect AI-generated noise.
- Analyze pixel patterns through various upsampling layers.
- Use texture-based analysis to focus on micro-patterns.
- Combine spatial, frequency, and semantic cues for a versatile detection model.

Why this matters for DeepFakeDetector

- **Use FFT (Fast Fourier Transform - converts pixels to frequency domain) analysis** to find subtle frequency artifacts that pixel models miss.
- **Build a hybrid model** combining CNNs (for pixel patterns), ViTs (for overall context), and FFT features (for frequency clues).
- **Add noise and texture checks** to detect missing camera noise or overly smooth surfaces.
- **Train on a mix of real and AI images** from many generators to make the detector adaptable.